

Matrix inequalities and norms

A sum-of-squares representation for the Toeplitz commutator form

Difficulty: challenging **Importance:** interesting to specialist**Status:** Partially resolved**Provenance:** explicit conjecture**Rating rationale:** All-order algebraic certification needs more than known nonnegativity; its direct application is a specific Toeplitz commutator form.

Finite-order SOS certificates — 12 September 2026

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Sections 1–5 of the [proof](#) establish sum-of-squares representations for orders $n = 8, 9, 10, 11, 12$, with at most $2(n-1)^2$ homogeneous quadratic squares. The five rational Gram certificates passed exact polynomial-identity and integer positive-definiteness checks. The assertion for every order remains open.

A separate [independent Codex AI-agent audit](#) passed this limited scope. This is informal automated review; no complete resolution, historical novelty, external human peer review or formal verification is claimed. No Lean verification was performed.

Problem statement

For an integer $n \geq 2$, let $x_{-(n-1)}, \dots, x_{n-1}, y_{-(n-1)}, \dots, y_{n-1}$ be independent real variables. Form real Toeplitz matrices $X = (x_{i-j})_{i,j=1}^n$ and $Y = (y_{i-j})_{i,j=1}^n$, and the homogeneous quartic polynomial

$$F_n(x, y) = 2\|X\|_F^2\|Y\|_F^2 - 2(\operatorname{tr}(X^T Y))^2 - \|XY - YX\|_F^2,$$

where $\|Z\|_F^2 = \sum_{i,j} z_{ij}^2$ for real matrices. Is it true that for every $n \geq 2$ there are a finite integer $N_n \geq 0$ and homogeneous quadratic polynomials $q_{n,1}, \dots, q_{n,N_n}$ with real coefficients in these $4n-2$ variables such that

$$F_n(x, y) = \sum_{j=1}^{N_n} q_{n,j}(x, y)^2$$

as a polynomial identity? The polynomials may depend on n . Rational coefficients are not required.

Relevance

An explicit sum-of-squares identity would give an algebraic certificate for a structured matrix inequality and connects Toeplitz calculations to semidefinite optimization. Nonnegativity alone does not answer this question.

References

1. L. László, *Sum of squares representation for the Böttcher–Wenzel biquadratic form*, Acta Universitatis Sapientiae, Informatica 4(1) (2012), 17–32: equation (1), §5, and Conjecture 15, p.31. [Primary manuscript.](#)
2. J. Ge, F. Li, Z. Tang, and Y. Zhou, *A survey on the DDVV-type inequalities*, Advances in Mathematics (China) 53 (2024), 449–467: published Conjecture 4.1, p.461; Conjecture 4.3 in [arXiv:2402.01085v1](#). [Published PDF.](#)

Status check — 2026-09-10

Both arXiv records remain v1; the published 2024 survey still poses this SOS assertion. László's particular certificate strategy works through order seven; its failure in a larger order is not a disproof of all SOS representations. Searches used *Toeplitz SOS Wenzel*, *Toeplitz Bottcher squares proof*, and *Toeplitz Conjecture 15 squares*. No general proof or non-SOS counterexample was located. The already proved Böttcher–Wenzel nonnegativity statement is not being counted again. The optional rational-certificate request in the original is not imposed as an additional conjecture.

Audit update (2026-09-10): Rechecked the survey's Conjecture 4.3 and searched for later Toeplitz SOS results. The general assertion remains a conjecture; known small-order certificates do not establish all-order representability. This is a bounded literature check, not a proof that no solution exists.